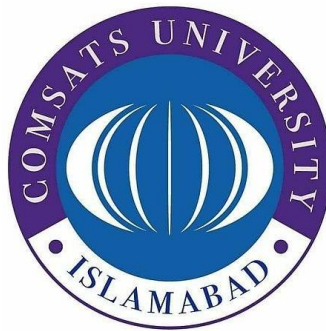


SmartTrade AI Automation System

Final Year Project Proposal Session

2023-2027

A project submitted in partial fulfilment of the
COMSATS University Degree
of
BS in Computer Science / Software Engineering (CUI)



Department of Computer Science
COMSATS University Islamabad, Lahore Campus

Project Registration

Project ID (for office use)					
Type of project	☐ Traditional ☐ Industrial ☐ Continuing				
Nature of project	☐ Development ☐ Research & Development				
Sustainable Development Goals (SDGs)	☐ Good Health and Well-Being ☐ Quality Education ☐ Industry, Innovation, and Infrastructure ☐ Gender Equality ☐ Decent Work and Economic Growth ☐ Climate Action				
Area of specialization	☐ Artificial Intelligence (AI) ☐ Blockchain ☐ Cybersecurity ☐ Data Science and Analytics ☐ Game Development ☐ Internet of Things (IoT) ☐ Natural Language Processing (NLP) ☐ Mobile App Development ☐ Web Development				
Project Group Members					
Sr.#	Reg. #	Student Name	Email ID	Phone #	Signature
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(iii)					
Declaration: FYP group members have cleared all prerequisite courses for FYP-I as per their degree requirements. For BS (Computer Science) and BS (Software Engineering) (CSC241 Object Oriented Programming, CSC291 Software Engineering Concepts, CSC270 Database Systems, HUM102 Report Writing Skills) For BS (Artificial Intelligence) (CSC291 Software Engineering Concepts, HUM102 Report Writing Skills)					

Plagiarism Free Certificate

This is to certify that, I am Mubashir Ashraf S/O Ashraf, group leader of FYP under registration no CUI /SP23-BSE-098/LHR at the Computer Science Department, COMSATS University Islamabad, Lahore Campus. I declare that my FYP proposal is checked by my supervisor and the similarity index is _____% that is less than 20%, an acceptable limit by HEC. The report is attached herewith as Appendix A.

Date: _____ Name of Group Leader: Mubashir Ashraf Signature: _____

Name of Supervisor: _____ Co-Supervisor (if any): _____

Designation: _____ Designation: _____

Signature: _____ Signature: _____

ABSTRACT

The quick growth of digital asset trading has brought millions of traders into the markets, but to be successful now, you need more than just manually executing trades—you need automation. Traders put in a lot of time and money to make indicators, expert advisors, and backtesting systems, but making reliable trading bots, indicators, and backtesting systems is still hard and expensive. Our solution changes the game by letting traders easily turn their strategies into Pine Script or MQL4/5, deploy fully automated trading bots, and run smart backtest with parameter optimization. This is a powerful, all-in-one platform that makes advanced trading automation easy, profitable, and available to all traders.

Our system is using Pine Script in Trading View for strategies. Also, we are removing the gap between outputs and Expert Adviser by using Python webhook and signal server in MT4. Basic features are position sizing, stop loss/take profit enforcement, drawdown monitoring and recording all the trades executed. So, we remove the gap between the real time execution and those conceptual models by providing a robust pipeline. This solution proposed by us is solving problems like delayed decisions by humans and inability of non-technical traders to test their complicated strategies.

1. Introduction

During these last years, the continuous expanding nature of the virtual assets (stocks) has attracted hundreds of thousands of traders to this market. But due to fast paced conditions of markets nowadays doing work manually cannot guarantee your success rather it increases your effort and wastes your time. So, now in order to deal with this situation there are automation tools which have come in the market which are able to respond quickly to changes or fluctuations in prices leaving the inability of human beings to make delayed decisions due to emotional weaknesses. So, AI automation has revolutionized the way people are trading these days.

As we discussed above, market prices are fluctuating every time whether it is day or night. So, it is impossible for any single person no matter how much expert he/she is, cannot keep up with the market trends. That makes with manual trading it is easy to miss opportunities or lose money on errors. In order to be ahead of the pack, traders must have the automation software to assist them in making smarter decisions and react quicker without having to do it all manually every time.

This movement towards the automation has changed a lot of things. Like how people used to trade and how they used to formulate strategies and then execute it by themselves. As people who are connected to this trading business know that most of the profits from trading are earned through indicators. Yet it is very hard to make good and stable trading bots because it requires a lot of effort and cost. Effort in a sense that first a trader who has new strategy, first he has to find a good developer with good knowledge of math and business and on his own field which is obviously good grip on software development. And cost in a sense that it is a very expensive work as we want to build a stable trading strategies which requires extensive back testing.

2. Problem Statement

Most traders still employ manual means to execute trades and to backtest their strategies in the fast-changing universe of digital trading today. Manual trading consumes much time, is error-prone, and is often too slow to suit modern markets where prices change so quickly. The complexity of programming and the steep development cost also make it difficult for traders to develop reliable trading bots, indicators, and backtesting tools.

As a result, even skilled traders struggle to turn their ideas into automated systems that can work

efficiently and continuously. So, if automation is not used it creates gap between strategy design and real time execution. This gap can cause many lost opportunities in the market.

3. Proposed Solution

Basically, our projects present a fast and robust trading platform which is automated in such a way it is very convenient and friendly to use for a new user and this can streamline the whole process to address the issue discussed in problem statement. The best part about our solution is that our platform for someone who has no prior experience in programming has no need to worry at all. Because at our system, they can seamlessly make their strategies and use them in Pine Scripts or MQL4/5 code. We will provide traders with guidelines for how to manage risks and how to optimize the parameters in such a way that it brings a better outcome and then they can easily back test their theories using historical data. Now our main concern is to enable traders to focus mainly on strategy development rather than getting involved in technical complexity.

4. Research Gaps and Technological Opportunities

Nowadays, everything is getting so fast that you cannot keep up if you are still stuck with those primitive methods. Same is the case in trading world. There are some traders who still trade manually which is very time consuming and is open to possible errors which can cause a lot of risks when trading with huge sum of money. Also not to mention about how much time they spend while adjusting different parameters and making trades manually.

As most of traders are not enough introduced to software or AI related stuff so it is kind of hard for them to grasp technical terms and concepts rather than just focusing on trading strategies. Using new technologies and stuff to tackle with these kinds of problems is a great opportunity. Now you can easily develop or come up with a very robust, secure and fast workflow using Pine Script to make new strategies and Meta Trade to execute them.

5. Objectives

The main objective of this system will enable traders to develop, test, and run their trading strategies without requiring extensive technical expertise. An additional objective is to provide functionalities such as position sizing, stop-loss and take-profit levels, and automated trade recording for improved risk management.

Other aim is to bring sophisticated trading automation within reach of a broader segment of traders, ranging from novices to professionals. The system will focus on simplicity, affordability, and compatibility with different trading conditions. To help traders evaluate and refine their strategies before live trading, it also aims to make it easier to perform backtesting based on historical data.

6. Related work

1. Artificial Intelligence in Financial Trading Systems

These new researches in AI have most likely altered the way people used to develop trading software. A comprehensive review by [1] emphasizes the advancement and trend of the subject by systematically examining various AI methodologies used in financial trading [2].

2. Frameworks for Algorithmic Trading

There have been tons of studies which were conducted to optimize the trading algorithms. [2] [3], for example, both of these presents an empirical analysis of trading systems developed through

different approaches, dividing them into three types and measuring their efficiency.

3. Hybrid Architectures in Trading Systems

It has been proved that if we inject a machine learning model in a rule-based system then it is believed that user will get stable trading strategies from it. A paper by AI [4] describes the design of a hybrid algorithmic trading system integrating machine learning models and rule-based risk management.

4. Backtesting and Trading Platforms

Back Trader [5] is a popular Python library used to backtest trading strategies with ease of development and testing. [Trading View](#) [6] gives sophisticated charting capabilities and Pine Script to automate strategies. MT4/5 platforms are most common for forex and cryptocurrency automated trading. [YouTube videos and StatMetaDers](#) materials give step-by-step guides to developing and implementing automated trading systems.

7. Proposed Methodology and Architecture

7.1 Methodology

Requirement Elicitation & Analysis:

In this step we will gather requirements from traders, developers and experts in algorithmic trading to identify the functional and non-functional requirements for our platform. We will gather requirements by join online trading forums and Subreddits. Requirements for features like strategy-to-code conversion, automated bot development, Expert Advisors development, parameter-optimized backtesting, multi-exchange support, trading view charting interface integrations and risk management tools are most important for us.

System Design:

Designing the UI/UX mockups for the web and desktop. Designing backend architecture and workflow for whole system, database schema, and data flow for trading signals, market data, and bot execution are defined. Integration points with trading APIs (e.g., Meta trader, Trading view) are planned to ensure smooth automation.

Strategy Conversion Module Development:

This Module will be developed to translate user-defined trading strategies in natural language into algorithmic scripting languages like pine script for trading view and mql4/5 for the Meta trader bases on the requirement. This module will also have coding chat-bot that can help user for onwards difficulties like syntax validation, error handling and making changes in code according to their requirements. strategy templates are implemented to simplify more for translating strategy into needed coding languages,

Trading Bot Development:

Fully automated trading bots are built to execute strategies across supported exchanges. Features like order placement, opening and closing trades, stop-loss/limit order management, and real-time monitoring and journaling all trades. Both will be optimized for low latency and reliability to ensure trades are executed efficiently and effectively according to the traders need.

Backtesting & Optimization Engine:

Creating the backtesting module to simulate trading strategies on historical market data and on different timeframes. Parameter optimization algorithms are used to identify the best-performing parameters or configurations in each backtest by the backtesting engines. Results are visualized through interactive charts, performance metrics, and trade logs for informed decision-making.

Market Data Integration:

Fetching the Real-time market data from different sources using secure APIs like finance, interactive broker. Data likes price feeds, order book depth, and historical OHLCV. Proper and secure coaching and rate-limit handling ensure system stability during high-frequency updates.

Risk Management & Alerts:

it includes the risk management features like maximum draw limits, position sizing and volatility filters, are implemented. Notification alerts system to notify users for significant email, push notifications, and desktop alerts.

Validation and Testing:

integration testing, system testing, Unit testing are performed to ensure the platform's reliability and trader assets safety. Backtesting results are validated against real historical data to confirm accuracy. Beta testing with selected traders ensures usability and performance in live-like conditions.

Deployment and Maintenance:

Deploying the platform to cloud servers for security, high availability and scalability. Continuous monitoring, updates and support are provided to main trading bots and expert advisors performance, add new features and adapt to continuous changing market conditions and technology change.

7.2 System Architecture

1. Presentation Layer

- This layer includes chatbot interface where traders can describe and write their strategies in plain language.
- A dashboard displays important information like backtesting results, optimized parameters, and performance of live bots.
- It will also provide interactive trade charts for trade history, signal visualization, and market analysis, allowing users to easily track how their strategies behave over time.

2. Chatbot and Strategy Generation Layer:

- This layer will understand and translate the user natural language.
- Here chatbot interprets natural language instructions and converts them into structured commands. Based upon those, it will automatically generate strategy code in user required platform formats like MQL4/5 for Meta trader, Pine script for Trading View, or Python using Back trader /backtesting.py.
- This layer also checks its generated code for syntax or logical issues before actual execution. In addition, it will also suggest improvements and recommend parameter adjustments based on previous results and strategy patterns.

3. Back testing & Optimization Layer

This layer runs multiple backtest across different timeframes and historical datasets to evaluate each strategy. Basically, what it does that it goes searching for such combinations of parameters which can maximize the performance while managing the whole risk and controlling the aim for good returns. After the back testing, all the results are summarized and then recorded in metrices and then these metrices are sent back to AI module for reinforced learning and to presentation layer.

4. Trading Execution Layer:

After the strategy is tested and later optimized then this layer (TEL) is responsible for running this strategy in real markets. It also handles live monitoring, error handling and basic risk controls to make sure strategies operate reliably during its execution

5. Market Data Integration Layer

This layer helps system to get the real-time and historical market data. It is connected to the data sources like Trading view and MT4/5 APIs for fetching prices, indicators, and other relevant information. Also, in this layer data is cleaned, standardized and stored in a specific and suitable

format for both backtesting and live trading, Caching mechanism is also used to reduce latency for frequently accessed data.

6. Data & Storage Layer

Here in this layer different databases will be used like PostgreSQL store structured data and Redis is used for caching. A Pinecone vector database is used for RAG (retrieval-augmented generation), helps AI to refer back to previous strategies and performance results, metrics to make better suggestions over time.

Architecture Diagram

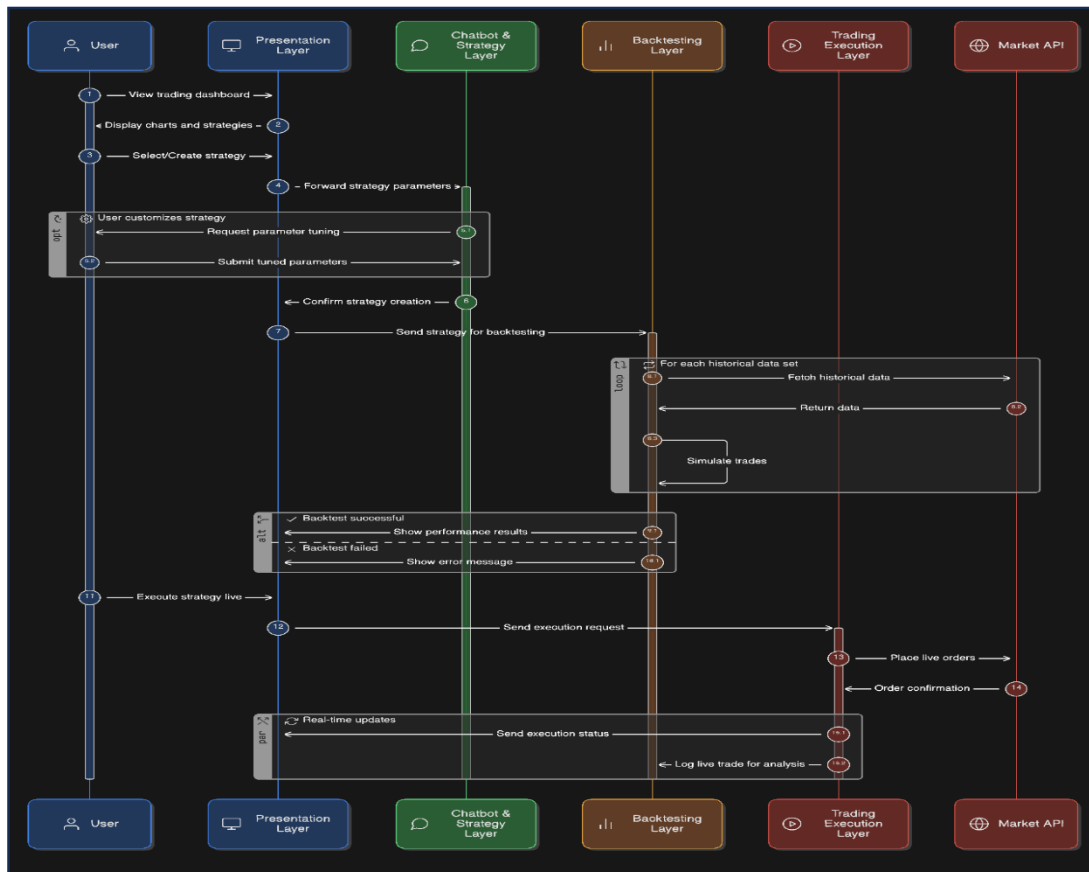
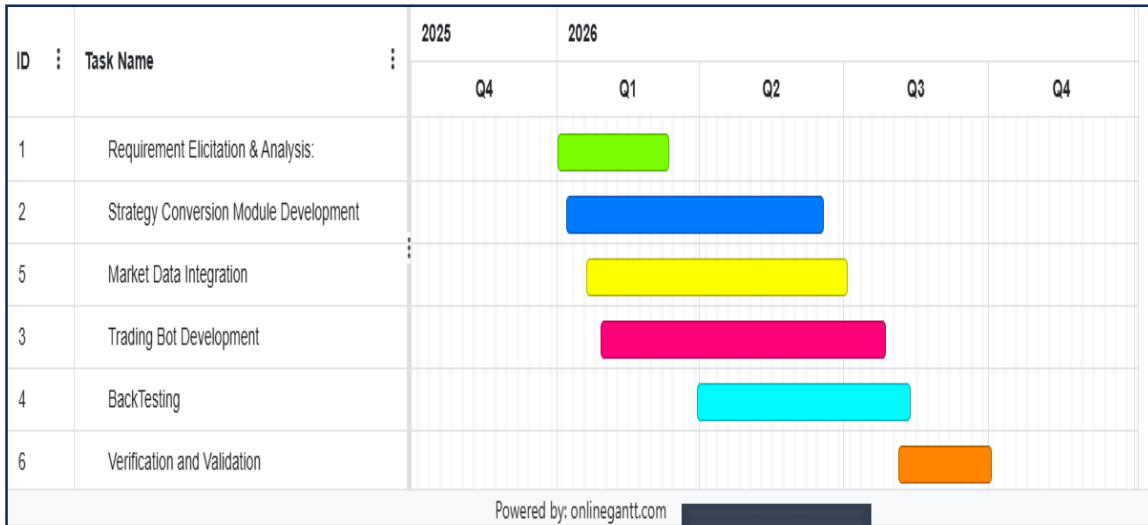


Figure 1: System Architecture Diagram

Ghant chart



8. Success Criterion

Criterion	Test Method	Expected Result / Target
<i>Data Integration Accuracy</i>	Test data fetching from APIs such as Trading View and Meta Trader 4 (MT4) for different instruments.	≥ 95% correct synchronization of live and historical data without data loss or delay.
<i>Strategy Execution Reliability</i>	Run automated strategies based on technical indicators and patterns under real-time conditions.	≥ 90% of trade signals executed correctly according to defined rules and market conditions.
<i>Backtesting Performance</i>	Execute backtesting on multiple datasets and compare with real-time outcomes.	Backtesting results ≥ 85% aligned with live strategy performance, ensuring reliability.
<i>System Latency & Execution Speed</i>	Measure time from signal generation to trade execution.	≤ 1.5 seconds total latency under normal network conditions.
<i>Risk Management Accuracy</i>	Simulate stop-loss, take-profit, and position-sizing scenarios.	100% correct risk control actions triggered according to predefined thresholds.
<i>Security & API Compliance</i>	Verify secure API key handling, encryption, and user authentication.	100% compliance with secure authentication and no data exposure.
<i>System Scalability</i>	Test system performance with	No significant slowdown or failure

Criterion	Test Method	Expected Result / Target
	increased trading pairs and simultaneous users.	up to 200 concurrent strategy instances.

9. Team Roles and Responsibilities

Team Member	Role	Tasks	Tentative Completion
Mubashar Ashraf	AI & Full-Stack Engineer	- Jointly develop frontend & backend modules (chatbot, dashboard, APIs). - Integrate AI (LLMs, Lang Chain/Lang Graph, RAG) for strategy generation & optimization. - Work on backtesting, Dockerization, deployment, and testing.	Month 1–7
Fahad Aziz	Full-Stack & Deployment Engineer	- Co-develop frontend/backend with real-time AI integration. - Implement backtesting workflows & Trading View visualization. - Contribute to architecture, containerization, CI/CD, and system validation.	Month 1–8

10. Tools and Technologies

- **Lang:** Python, JavaScript, Pine script, Mql4/5, back trader script
- **LLMs:** Claude, OpenAI, DeepSeek, Gemini
- **Frameworks:** Lang chain, Lang Graph, back trader, react.js
- **Database:** MongoDB, pinecone
- **Platforms:** trading view, Metatrader4, Meta trader 5
- **Deployment:** Azure Cloud, Docker, Wine

11. References

- [1] F. Dakalbab, Science direct.
- [2] T. &. Frances, "Automated Trading System," Huang B., 2019.
- [3]
- [4] "Architecture of Trading Algorithm," Medium, 2025.
- [5] "Back Trader," 2024.
- [6] "Trading view and chart pine".
- [7] ScienceDirect.

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